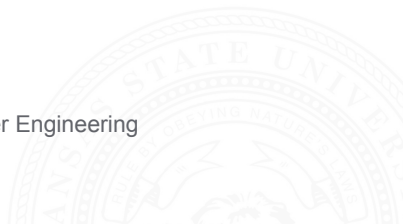


Transformer Modeling II

From winding physics to a checked operating-point model

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Lecture outline

- 1 Build the practical transformer model from physical effects
- 2 Refer the circuit and justify the input-shunt approximation
- 3 Solve a complete load example using the chosen circuit model

Voltage and current phasor magnitudes are RMS; referring a phasor across an ideal transformer does not change that convention.

PART 1

Practical transformer model

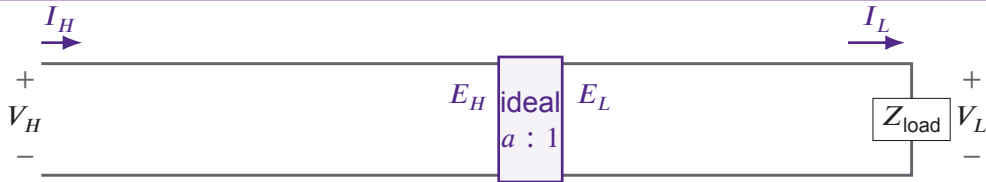
Winding effects, core excitation, and the exact equivalent circuit

Ideal-model limitation

STARTING PROBLEM

A loaded transformer has the correct turns ratio, yet its terminal voltage is below the ideal value and its input power exceeds its output power. Which physical effects must a practical model add?

Ideal transformer equivalent circuit



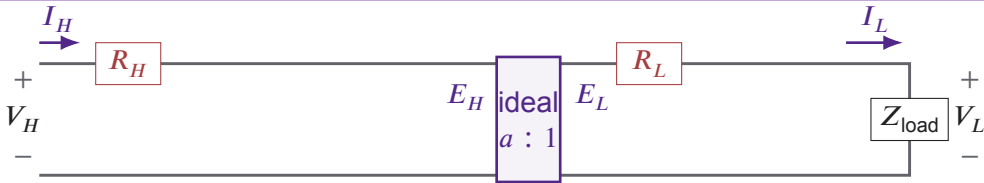
The ideal element transfers power between two electrically separate circuits.

$$a = \frac{N_H}{N_L}, \quad E_H = aE_L, \quad I_H = \frac{I_L}{a}$$

With no winding voltage drops, $V_H = E_H$ and $V_L = E_L$.

The ideal ratio element remains as the physical effects are added around it.

Winding resistance



Ideal assumption

Zero winding resistance:

$$R_H = R_L = 0.$$

Physical transformer

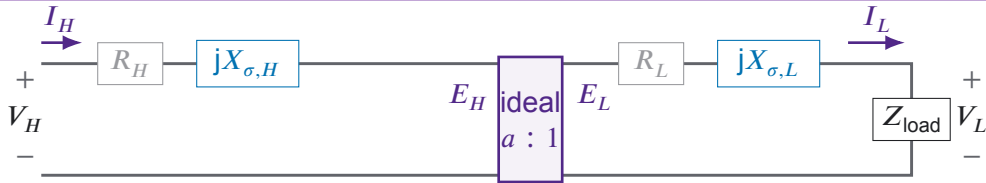
Finite resistance causes voltage drop and heating.

Series elements: R_H and R_L .

$$\Delta V_R = IR, \quad P_{cu} = |I_H|^2 R_H + |I_L|^2 R_L$$

Winding resistance causes an in-phase voltage drop and real copper loss.

Leakage flux



Ideal assumption

All flux links both windings:

$$X_{\sigma,H} = X_{\sigma,L} = 0.$$

Physical transformer

Some flux links only one winding.

Add $jX_{\sigma,H}$ and $jX_{\sigma,L}$ in series.

$$\Delta V_{\sigma} = jX_{\sigma}I, \quad X_{\sigma} = \omega L_{\sigma}$$

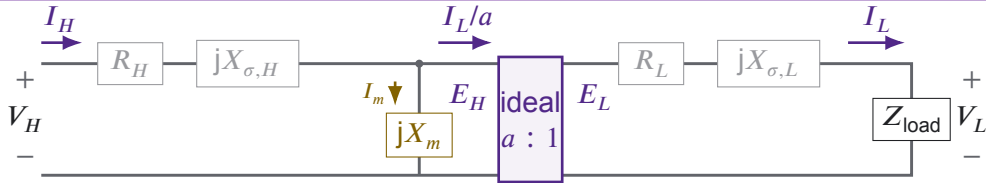
Leakage reactance adds a voltage drop 90° ahead of winding current.

Why resistance and leakage reactance?

	Resistance R	Leakage reactance jX_σ
Origin	Finite winding conductivity	Leakage flux
Phase	ΔV_R is in phase with I	ΔV_σ leads I by 90°
Energy	Converted to heat	Stored in the field and returned

R models heating; jX_σ models magnetic energy storage.

Finite core permeability



Ideal assumption

Infinite permeability:
no magnetizing current is required.

Physical transformer

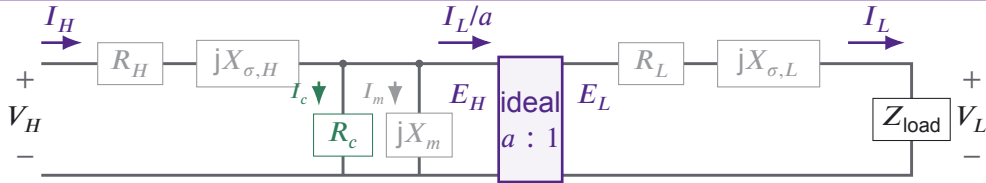
Finite permeability requires magnetizing current.

I_m lags E_H by 90° .

$$I_m = \frac{E_H}{jX_m}, \quad I_H = \frac{I_L}{a} + I_m$$

At no load, E_H still drives magnetizing current; jX_m is a shunt branch.

Core loss



Ideal assumption

A lossless core:
no real core power is absorbed.

Physical transformer

Hysteresis and eddy currents absorb real power.
 I_c is in phase with E_H .

$$I_c = \frac{E_H}{R_c}, \quad P_{\text{core}} = \frac{|E_H|^2}{R_c}$$

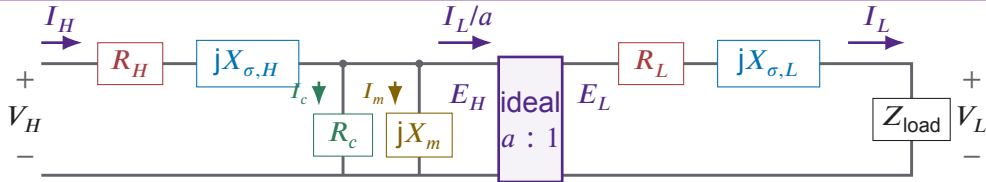
Core loss persists at no load; R_c draws current from E_H in parallel with jX_m .

Why R_c and jX_m ?

	Core loss: R_c	Magnetization: jX_m
Origin	Hysteresis and eddy currents	Finite core permeability
Phase	I_c is in phase with E_H	I_m lags E_H by 90°
Energy	Converted to heat	Stored in the field and returned

Both are shunt branches across E_H : the core remains excited at no load.

Exact equivalent circuit



Input current supplies the load and both excitation branches.

$$I_0 = I_c + I_m, \quad I_H = \frac{I_L}{a} + I_0$$

$$V_H = E_H + I_H(R_H + jX_{\sigma,H}) \quad E_L = V_L + I_L(R_L + jX_{\sigma,L})$$

The internal ratio remains $E_H = aE_L$; winding drops change the terminal voltages.

Source: Glover et al., 7th ed., Ch. 3. Course redraw; linear sinusoidal steady-state model.

PART 2

Referral and approximation

Impedance referral, excitation current, and the input-shunt approximation

Why referral and approximation?

We want terminal voltages and currents with less algebra.

Referral

Refer both windings to one side so we can solve a single circuit.

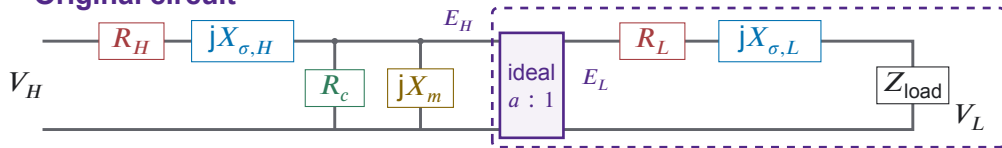
Approximation

Simplify effects we have checked are small, making voltage-drop calculations easier.

Referral is exact. Approximation trades some accuracy for simpler calculations.

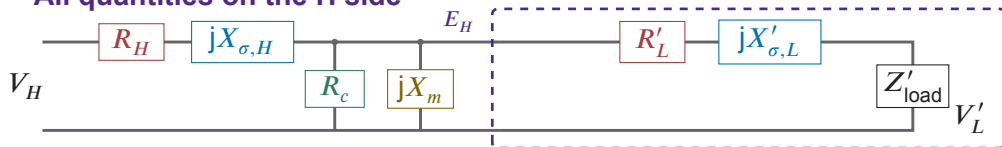
Referral: replace the boxed network

Original circuit



Same behavior seen at E_H ↓

All quantities on the H side



The replacement is exact. We can now analyze the whole circuit on the H side.

Refer low-side quantities to H

With $a = N_H/N_L$, define the quantities referred to the H side:

Voltage

$$V'_L = aV_L$$

Current

$$I'_L = \frac{I_L}{a}$$

A prime marks a referred value. These definitions let us calculate $Z'_L = V'_L/I'_L$.

In general, $V'_L \neq V_H$: winding impedances still cause voltage drops.

Referral derivation

$$Z'_L = \frac{V'_L}{I'_L} = \frac{aV_L}{I_L/a} = \boxed{a^2 Z_L}.$$

Winding

$$Z'_{w,L} = a^2 Z_{w,L}$$

Refer both resistance and leakage reactance.

Load

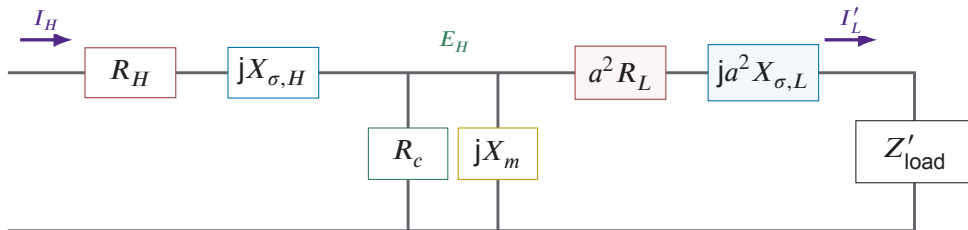
$$Z'_{\text{load}} = a^2 Z_{\text{load}}$$

The load must move to the same side as the circuit.

Voltage uses a , current uses $1/a$, and impedance uses a^2 .

Equivalent circuit referred to H

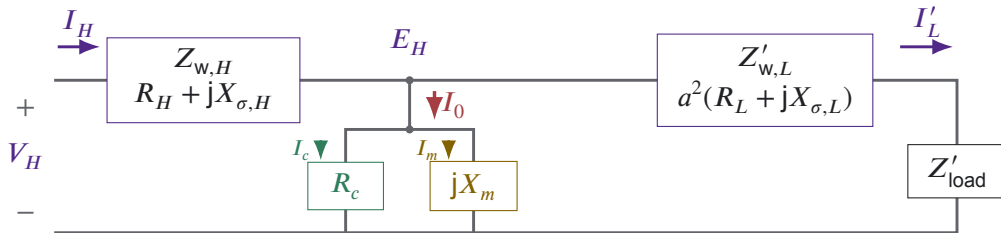
Exact circuit on the high-voltage side



L to H: multiply low-side impedances by a^2

Referral changes numerical values while preserving terminal behavior and complex power.

Excitation current



$$I_0 = I_H - I'_L = I_c + I_m$$

$$I_0 = E_H \left(\frac{1}{R_c} + \frac{1}{jX_m} \right)$$

I_0 is the phasor gap between I_H and I'_L .

Why the excitation branch can move

At normal loaded operation:

Large R_c and X_m : small total excitation current

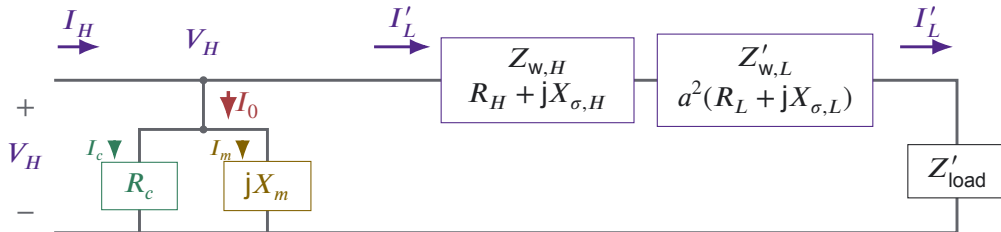
$$|I_0| \ll |I'_L| \quad \Rightarrow \quad I_H \approx I'_L$$

Small winding impedance: small voltage drop

$$|V_H - E_H| \ll |V_H| \quad \Rightarrow \quad E_H \approx V_H$$

Moving $R_c \parallel jX_m$ to V_H changes excitation and winding voltage drop only slightly.

Input-shunt approximation



Evaluate excitation at V_H instead of E_H .

$$I_0 \approx V_H \left(\frac{1}{R_c} + \frac{1}{jX_m} \right), \quad I_H = I'_L + I_0$$

Keep both shunt elements. Both winding impedances now carry I'_L .

Equivalent series impedance

In the circuit just drawn, both winding impedances carry the same current I'_L .

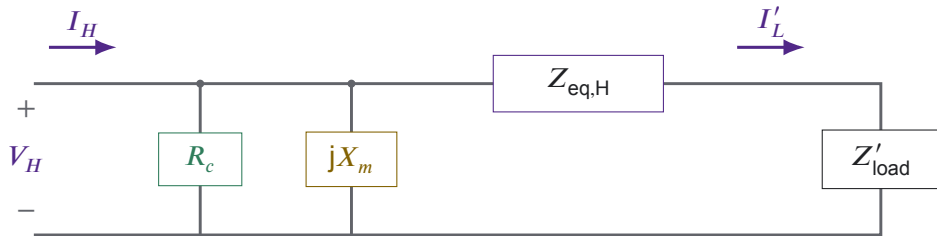
$$V_H - V'_L = I'_L Z_{w,H} + I'_L Z'_{w,L} = I'_L (Z_{w,H} + Z'_{w,L}).$$

$$Z_{eq,H} = Z_{w,H} + Z'_{w,L} = R_{eq,H} + jX_{eq,H}$$

$$R_{eq,H} = R_H + a^2 R_L, \quad X_{eq,H} = X_{\sigma,H} + a^2 X_{\sigma,L}$$

The windings share I'_L and can be combined; excitation current remains in the shunt branch.

Approximate circuit with excitation

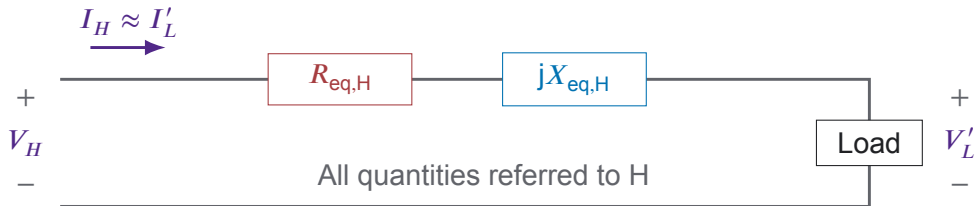


This input-shunt model retains excitation. Further simplifications follow for power-system studies.

$$I'_L = \frac{V_H}{Z_{eq,H} + Z'_{load}}, \quad I_0 = \frac{V_H}{R_c} + \frac{V_H}{jX_m}, \quad I_H = I'_L + I_0$$

Solve load current and excitation current separately, then add them as phasors.

Neglecting excitation current



When excitation current is small relative to load current, omit the shunt branch. Then

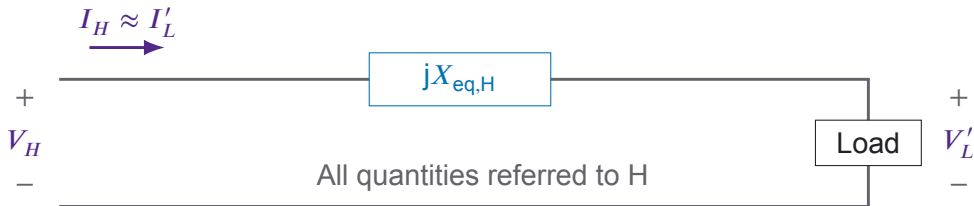
$$I_H \approx I'_L.$$

$$V_H \approx V'_L + I'_L(R_{eq,H} + jX_{eq,H})$$

Use this model for loaded voltage and current calculations. Retain the shunt branch when core loss or excitation matters.

Source: Glover et al., 6th ed., Sec. 3.2, p. 103, Fig. 3.6(b).

Neglecting winding resistance



If $R_{eq,H} \ll X_{eq,H}$ and winding loss is unimportant to the study, also omit $R_{eq,H}$.

$$V_H \approx V'_L + jX_{eq,H}I'_L$$

This further simplification is often useful for large power transformers. Keep resistance when its voltage drop or loss matters.

Source: Glover et al., 6th ed., Sec. 3.2, p. 103, Fig. 3.6(c).

Limits of the steady-state model

The equivalent circuit assumes sinusoidal steady state and constant parameters near the operating point.

- **Saturation:** core permeability changes with flux level, so a fixed X_m is insufficient.
- **Inrush:** energization can produce a large transient current that a steady-state phasor circuit cannot predict.
- **Excitation harmonics:** a nonlinear core draws nonsinusoidal current even with sinusoidal voltage.
- **Surges:** fast transients require winding capacitances absent from this circuit.

Choose a model that includes the physical effect being studied.

Source: Glover et al., 6th ed., Sec. 3.2, pp. 105–107.

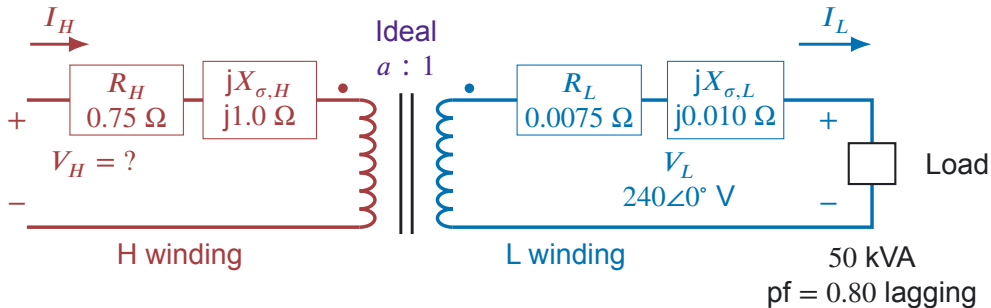
PART 3

Worked example

Model selection, referral, terminal quantities, and a circuit check

Example: data and required quantities

Single-phase transformer rated 50 kVA, 2400/240 V. All phasors are RMS.



Model: neglect excitation current. Retain both winding impedances.

Find: I_L , I_H , and the required V_H . Verify the result using KVL.

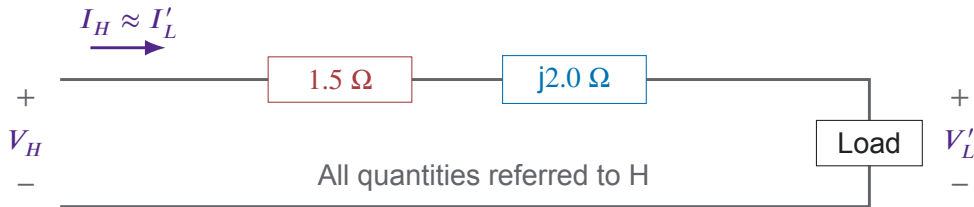
Source: Adapted from Glover et al., 6th ed., Problems 3.16–3.17(a,b), p. 146.

Example: circuit referred to H

With excitation neglected, refer the L winding to H and combine the two series impedances.

$$a = \frac{2400}{240} = 10, \quad Z'_{w,L} = 100(0.0075 + j0.010) = 0.75 + j1.0 \, \Omega$$

$$Z_{eq,H} = Z_{w,H} + Z'_{w,L} = (0.75 + j1.0) + (0.75 + j1.0) = \boxed{1.5 + j2.0 \, \Omega}$$



Neglect excitation current and retain both winding resistance and leakage reactance.

Example: load current

The load receives 50 kVA at 240 V. Its 0.80 lagging power factor fixes the current angle relative to V_L .

$$|I_L| = \frac{50,000}{240} = 208.333 \text{ A}, \quad \phi = \cos^{-1}(0.80) = 36.870^\circ$$

$$I_L = 208.333 \angle (-36.870^\circ) \text{ A}$$

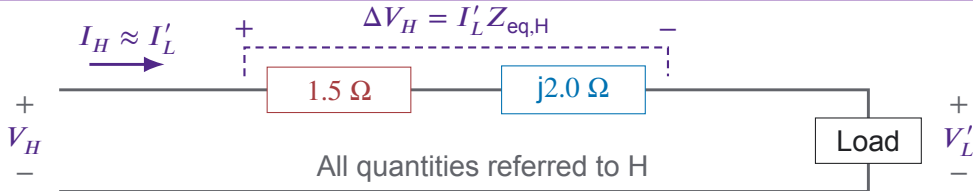
Refer the load voltage and current to H using the same $a = 10$:

$$V'_L = aV_L = 2400 \angle 0^\circ \text{ V}$$

$$I'_L = \frac{I_L}{a} = 20.833 \angle (-36.870^\circ) = 16.667 - j12.500 \text{ A}$$

The load voltage is the phase reference. The referred current lags it by the same 36.870° .

Example: required input voltage



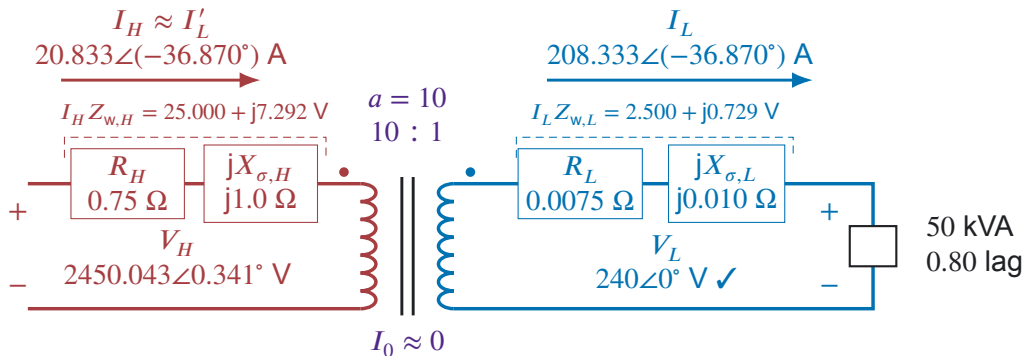
1. Calculate the winding voltage drop

$$\begin{aligned}\Delta V_H &= I'_L Z_{eq,H} = (16.667 - j12.500)(1.5 + j2.0) \\ &= 50.000 + j14.583 \text{ V}\end{aligned}$$

2. Add the drop to the referred load voltage

$$\begin{aligned}V_H &\approx V'_L + \Delta V_H = 2400 + (50.000 + j14.583) \\ &= 2450.000 + j14.583 = \boxed{2450.043 \angle 0.341^\circ \text{ V}}\end{aligned}$$

Example: results and circuit check



$$V'_L = V_H - I'_L Z_{eq,H} = (2450 + j14.583) - (50 + j14.583) = 2400 \text{ V}$$

$$V_L = V'_L / 10 = 240 \angle 0^\circ \text{ V} \checkmark$$